HOUSE PRICE PREDICTION MODEL

PROJECT REPORT

Submitted By:

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INTRODUCTION

House Price Prediction is an important application of Machine Learning that helps estimate the price of residential properties based on various features. Accurate house price prediction helps buyers, sellers, and real estate companies make informed decisions.

The real estate market contains many factors that influence property prices such as location, area, number of bedrooms, construction status, and availability of facilities. Machine Learning algorithms can analyze these factors and predict prices with good accuracy.

This project develops a House Price Prediction Model using the Random Forest Regressor algorithm.

Problem Statement: -

Determining the correct market value of a house is challenging because many factors affect the final price. The objective is to build a machine learning model that can predict house prices based on available property features.

OBJECTIVE AND DATASET DESCRIPTION

Project Objectives: -

- Predict house prices using Machine Learning.
- Perform data preprocessing and cleaning.
- Train and test a prediction model.
- Evaluate model performance using metrics.
- Visualize prediction results.

Dataset Description: -

The dataset contains information about residential properties.

Important attributes include:

- POSTED_BY
- UNDER_CONSTRUCTION
- RERA
- BHK_NO
- BHK_OR_RK
- SQUARE_FT
- READY_TO_MOVE
- RESALE
- LATITUDE
- LONGITUDE
- TARGET(PRICE_IN_LACS)

The target variable is: -

TARGET(PRICE_IN_LACS)

which represents the house price in lakhs.

METHODOLOGY

Data Preprocessing

The following preprocessing steps were performed:

1. Dataset loading using Pandas.
2. Feature selection.
3. Removal of unnecessary columns such as ADDRESS.
4. One-Hot Encoding of categorical variables.
5. Splitting data into training and testing sets.

Train-Test Split

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

This ensures proper evaluation of the model.

Machine Learning Model: -

Random Forest Regressor

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Advantages: -

- High accuracy
- Handles large datasets efficiently
- Works well with categorical and numerical data

IMPLEMENTATION AND RESULTS

Model Training: -

The Random Forest Regressor was trained using the training dataset.

Model Evaluation Metrics: -

The following evaluation metrics were used:

Mean Absolute Error (MAE)

Measures the average prediction error.

Root Mean Squared Error (RMSE)

Measures the standard deviation of prediction errors.

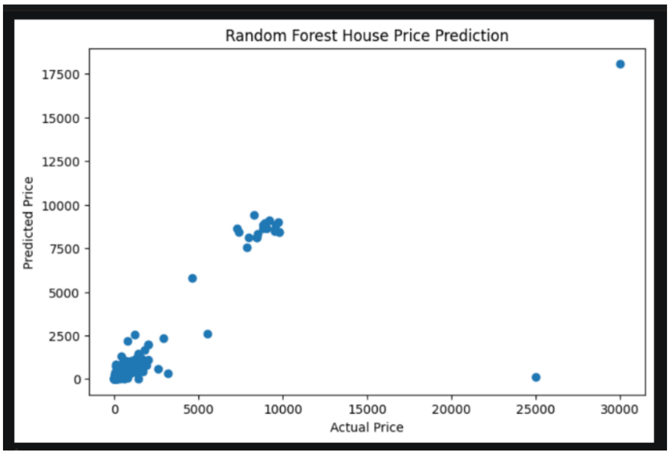
R^2 Score

Measures how well the model explains the variation in house prices.

Results: -

- MAE = 34.96
- RMSE = 374.03
- R^2 Score = 0.7432

The model achieved good prediction performance and explained approximately 74% of the variation in house prices.

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Actual Price = 34.9
Predicted Price = 63.57700000000002
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CONCLUSION AND FUTURE SCOPE

Conclusion: -

This project successfully developed a House Price Prediction Model using Random Forest Regressor. Data preprocessing, feature engineering, model training, and evaluation were performed successfully.

The model achieved an R^2 score of 0.7432, indicating good predictive capability.

Future Scope:-

- Hyperparameter tuning for improved accuracy.
- Use of advanced algorithms such as XGBoost and Gradient Boosting.
- Integration with real-time real estate platforms.
- Deployment as a web application for end users.

References: -

1. Python Documentation
2. Scikit-Learn Documentation
3. Pandas Documentation
4. Housing Dataset Source